

Problem 3: Favourite Algorithm vs Decision Tree (10-fold CV)

Method: paired t-test on per-fold differences (both algorithms evaluated on the same 10 folds). H_0 : mean difference = 0. H_1 : the favourite algorithm has the lower error.

Formulae

$$\bar{d} = (1/k) \sum d_i, s_d = \sqrt{(\sum (d_i - \bar{d})^2 / (k(k-1)))}, t = \bar{d} / s_d, k = 10, v = k - 1 = 9$$

Fold	Fav. alg. (%)	Dec. tree (%)	d_i = Fav - DT	(d_i - mean) ²
1	8.89	9.3	-0.41	0.0666
2	9.52	9.48	0.04	0.0369
3	8.13	9.12	-0.99	0.7022
4	9.48	9.13	0.35	0.2520
5	10.12	9.98	0.14	0.0853
6	10.23	11.01	-0.78	0.3944
7	8.56	9.02	-0.46	0.0949
8	9.12	8.56	0.56	0.5069
9	9.23	9.23	0.00	0.0231
10	9.11	9.08	0.03	0.0331
Sum			-1.52	2.1954

Calculation

$$\bar{d} = -1.52 / 10 = -0.152$$

$$\text{sample std dev of } d_i = \sqrt{(2.1954 / 9)} = 0.4939$$

$$s_d = 0.4939 / \sqrt{10} = 0.1562$$

$$t = -0.152 / 0.1562 = -0.97$$

Table 4 lookup (v = 9)

$$t_{65\%,9} = 0.985, t_{90\%,9} = 1.83, t_{95\%,9} = 2.26$$

$|t| = 0.97$ is just below 0.985, i.e. the two-sided confidence is just under 65%, so the one-sided confidence that the favourite algorithm is better is just under $(100+65)/2 = 82.5\%$. Exact t-distribution value: 82.2%.

Answer: The favourite algorithm has the lower mean error (by 0.152 percentage points), but we can assume it outperforms the decision tree with only about 80% confidence ($\approx 82\%$ by the table). This is well short of the 90% / 95% levels, so the difference is not statistically significant.